

2016 HIGHER SCHOOL CERTIFICATE EXAMINATION

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–13

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which sum is equal to $\sum_{k=1}^{20} (2k+1)$?

(A) $1 + 2 + 3 + 4 + \cdots + 20$

(B) $1 + 3 + 5 + 7 + \cdots + 41$

(C) $3 + 4 + 5 + 6 + \cdots + 20$

(D) $3 + 5 + 7 + 9 + \cdots + 41$

2 What is the remainder when $2x^3 - 10x^2 + 6x + 2$ is divided by $x - 2$?

(A) -66

(B) -10

(C) $-x^3 + 5x^2 - 3x - 1$

(D) $x^3 - 5x^2 + 3x + 1$

3 Which expression is equivalent to $\frac{\tan 2x - \tan x}{1 + \tan 2x \tan x}$?

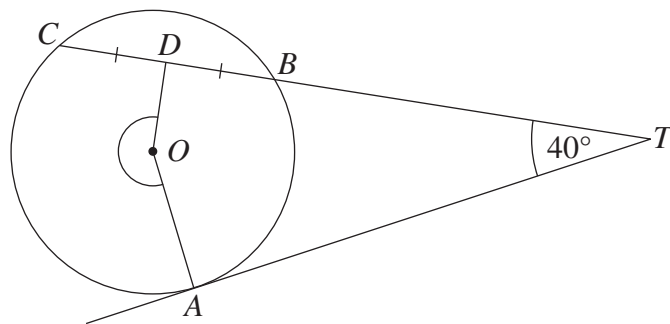
(A) $\tan x$

(B) $\tan 3x$

(C) $\frac{\tan 2x - 1}{1 + \tan 2x}$

(D) $\frac{\tan x}{1 + \tan 2x \tan x}$

- 4 In the diagram, O is the centre of the circle ABC , D is the midpoint of BC , AT is the tangent at A and $\angle ATB = 40^\circ$.



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What is the size of the reflex angle DOA ?

- (A) 80°
 (B) 140°
 (C) 220°
 (D) 280°
- 5 Which expression is equal to $\int \sin^2 2x \, dx$?

- (A) $\frac{1}{2} \left(x - \frac{1}{4} \sin 4x \right) + c$
 (B) $\frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c$
 (C) $\frac{\sin^3 2x}{6} + c$
 (D) $\frac{-\cos^3 2x}{6} + c$

6 What is the general solution of the equation $2\sin^2 x - 7\sin x + 3 = 0$?

(A) $n\pi - (-1)^n \frac{\pi}{3}$

(B) $n\pi + (-1)^n \frac{\pi}{3}$

(C) $n\pi - (-1)^n \frac{\pi}{6}$

(D) $n\pi + (-1)^n \frac{\pi}{6}$

7 The displacement x of a particle at time t is given by

$$x = 5\sin 4t + 12\cos 4t.$$

What is the maximum velocity of the particle?

(A) 13

(B) 28

(C) 52

(D) 68

8 A team of 11 students is to be formed from a group of 18 students. Among the 18 students are 3 students who are left-handed.

What is the number of possible teams containing at least 1 student who is left-handed?

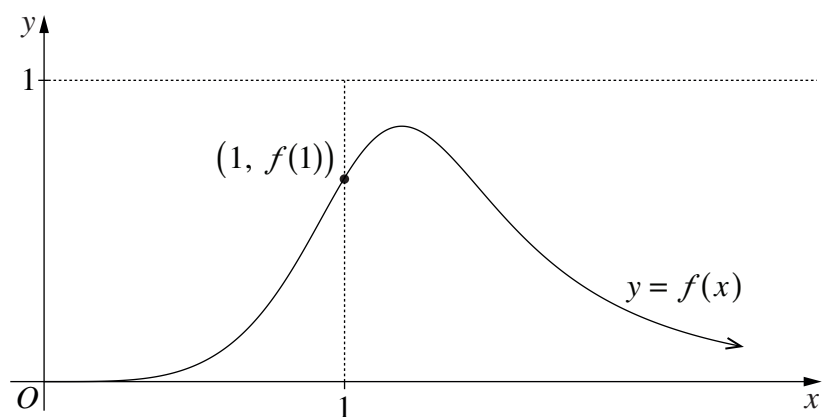
(A) 19 448

(B) 30 459

(C) 31 824

(D) 58 344

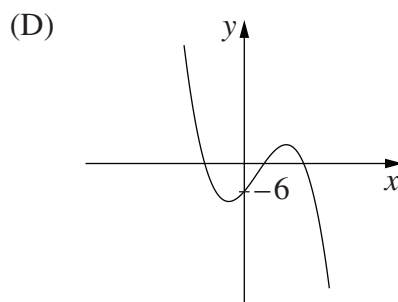
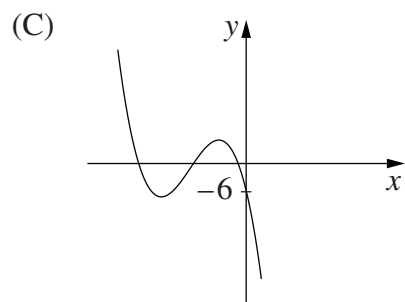
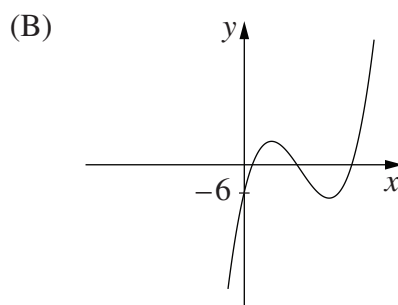
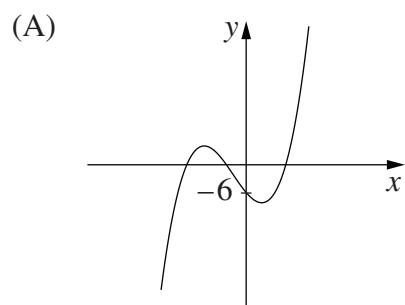
- 9 The diagram shows the graph of $y = f(x)$.



Which of the following is a correct statement?

- (A) $f''(1) < f(1) < 1 < f'(1)$
 (B) $f''(1) < f'(1) < f(1) < 1$
 (C) $f(1) < 1 < f'(1) < f''(1)$
 (D) $f'(1) < f(1) < 1 < f''(1)$
- 10 Consider the polynomial $p(x) = ax^3 + bx^2 + cx - 6$ with a and b positive.

Which graph could represent $p(x)$?



Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

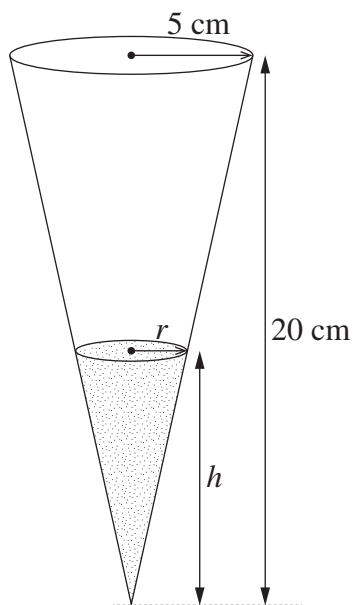
In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the inverse of the function $y = x^3 - 2$. **2**
- (b) Use the substitution $u = x - 4$ to find $\int x\sqrt{x-4} \, dx$. **3**
- (c) Differentiate $3\tan^{-1}(2x)$. **2**
- (d) Evaluate $\lim_{x \rightarrow 0} \left(\frac{2\sin x \cos x}{3x} \right)$. **2**
- (e) Solve $\frac{3}{2x+5} - x > 0$. **3**
- (f) A darts player calculates that when she aims for the bullseye the probability of her hitting the bullseye is $\frac{3}{5}$ with each throw.
- (i) Find the probability that she hits the bullseye with exactly one of her first three throws. **1**
- (ii) Find the probability that she hits the bullseye with at least two of her first six throws. **2**

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows a conical soap dispenser of radius 5 cm and height 20 cm.



At any time t seconds, the top surface of the soap in the container is a circle of radius r cm and its height is h cm.

The volume of the soap is given by $v = \frac{1}{3}\pi r^2 h$.

- (i) Explain why $r = \frac{h}{4}$. 1

- (ii) Show that $\frac{dv}{dh} = \frac{\pi}{16}h^2$. 1

The dispenser has a leak which causes soap to drip from the container. The area of the circle formed by the top surface of the soap is decreasing at a constant rate of $0.04 \text{ cm}^2 \text{ s}^{-1}$.

- (iii) Show that $\frac{dh}{dt} = \frac{-0.32}{\pi h}$. 2

- (iv) What is the rate of change of the volume of the soap, with respect to time, when $h = 10$? 2

Question 12 continues on page 8

Question 12 (continued)

- (b) In a chemical reaction, a compound X is formed from a compound Y . The mass in grams of X and Y are $x(t)$ and $y(t)$ respectively, where t is the time in seconds after the start of the chemical reaction.

Throughout the reaction the sum of the two masses is 500 g.

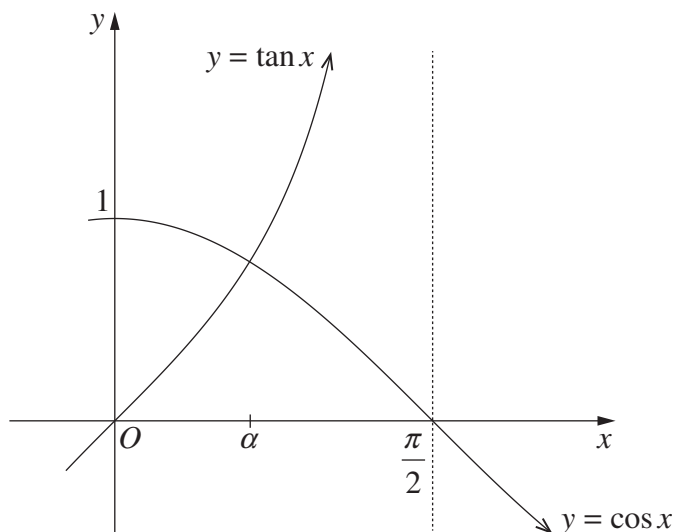
At any time t , the rate at which the mass of compound X is increasing is proportional to the mass of compound Y .

At the start of the chemical reaction, $x = 0$ and $\frac{dx}{dt} = 2$.

(i) Show that $\frac{dx}{dt} = 0.004(500 - x)$. **3**

(ii) Show that $x = 500 - Ae^{-0.004t}$ satisfies the equation in part (i), and find the value of A . **2**

- (c) The graphs of $y = \tan x$ and $y = \cos x$ meet at the point where $x = \alpha$, as shown.



(i) Show that the tangents to the curves at $x = \alpha$ are perpendicular. **2**

(ii) Use one application of Newton's method with $x_1 = 1$ to find an approximate value for α . Give your answer correct to two decimal places. **2**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The tide can be modelled using simple harmonic motion.

At a particular location, the high tide is 9 metres and the low tide is 1 metre.

At this location the tide completes 2 full periods every 25 hours.

Let t be the time in hours after the first high tide today.

- (i) Explain why the tide can be modelled by the function $x = 5 + 4\cos\left(\frac{4\pi}{25}t\right)$. **2**

- (ii) The first high tide tomorrow is at 2 am. **2**

What is the earliest time tomorrow at which the tide is increasing at the fastest rate?

Question 13 continues on page 10

Question 13 (continued)

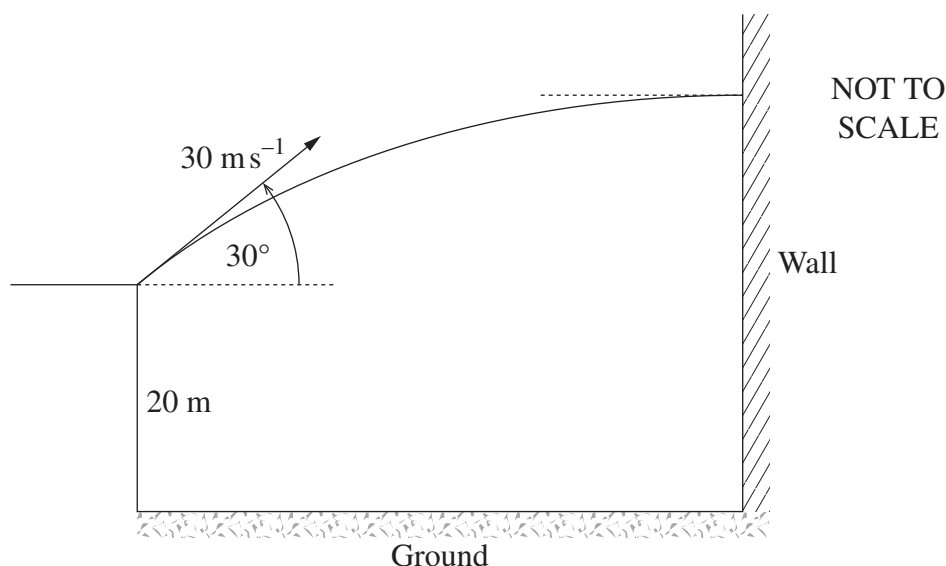
- (b) The trajectory of a projectile fired with speed $u \text{ m s}^{-1}$ at an angle θ to the horizontal is represented by the parametric equations

$$x = ut \cos \theta \quad \text{and} \quad y = ut \sin \theta - 5t^2,$$

where t is the time in seconds.

- (i) Prove that the greatest height reached by the projectile is $\frac{u^2 \sin^2 \theta}{20}$. 2

A ball is thrown from a point 20 m above the horizontal ground. It is thrown with speed 30 m s^{-1} at an angle of 30° to the horizontal. At its highest point the ball hits a wall, as shown in the diagram.



- (ii) Show that the ball hits the wall at a height of $\frac{125}{4}$ m above the ground. 2

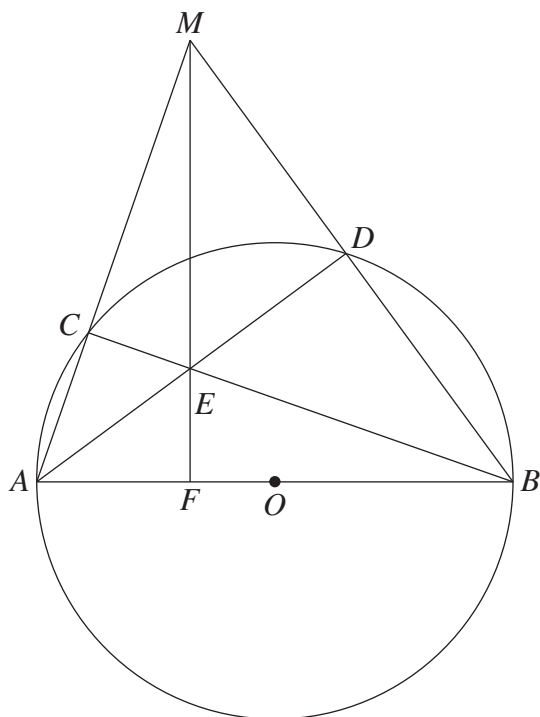
The ball then rebounds horizontally from the wall with speed 10 m s^{-1} . You may assume that the acceleration due to gravity is 10 m s^{-2} .

- (iii) How long does it take the ball to reach the ground after it rebounds from the wall? 2
- (iv) How far from the wall is the ball when it hits the ground? 1

Question 13 continues on page 11

Question 13 (continued)

- (c) The circle centred at O has a diameter AB . From the point M outside the circle the line segments MA and MB are drawn meeting the circle at C and D respectively, as shown in the diagram. The chords AD and BC meet at E . The line segment ME produced meets the diameter AB at F .



Copy or trace the diagram into your writing booklet.

- | | |
|--|----------|
| (i) Show that $CMDE$ is a cyclic quadrilateral. | 2 |
| (ii) Hence, or otherwise, prove that MF is perpendicular to AB . | 2 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that $4n^3 + 18n^2 + 23n + 9$ can be written as **1**

$$(n+1)(4n^2 + 14n + 9).$$

- (ii) Using the result in part (i), or otherwise, prove by mathematical induction that, for $n \geq 1$, **3**

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2n-1)(2n+1) = \frac{1}{3}n(4n^2 + 6n - 1).$$

- (b) Consider the expansion of $(1+x)^n$, where n is a positive integer.

- (i) Show that $2^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \cdots + \binom{n}{n}$. **1**

- (ii) Show that $n2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}$. **1**

- (iii) Hence, or otherwise, show that $\sum_{r=1}^n \binom{n}{r}(2r-n) = n$. **2**

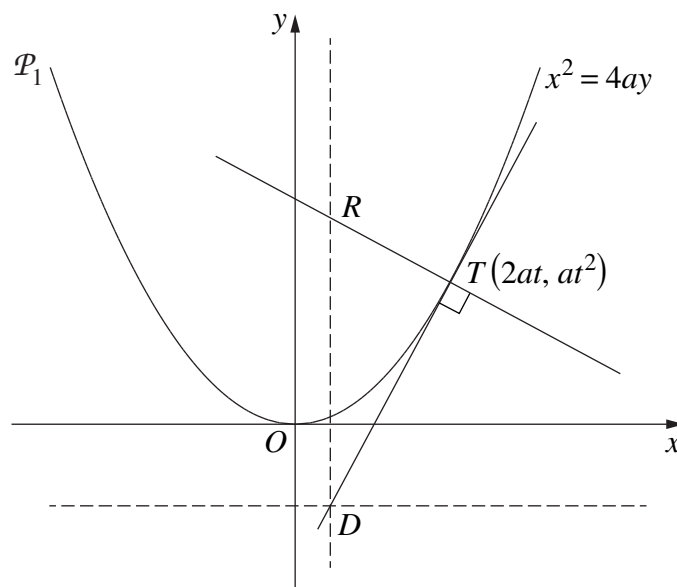
Question 14 continues on page 13

Question 14 (continued)

- (c) The point $T(2at, at^2)$ lies on the parabola $\mathcal{P}_1$ with equation $x^2 = 4ay$.

The tangent to the parabola $\mathcal{P}_1$ at T meets the directrix at D .

The normal to the parabola $\mathcal{P}_1$ at T meets the vertical line through D at the point R , as shown in the diagram.



- | | |
|---|----------|
| (i) Show that the point D has coordinates $\left(at - \frac{a}{t}, -a\right)$. | 1 |
| (ii) Show that the locus of R lies on another parabola $\mathcal{P}_2$. | 3 |
| (iii) State the focal length of the parabola $\mathcal{P}_2$. | 1 |

It can be shown that the minimum distance between R and T occurs when the normal to $\mathcal{P}_1$ at T is also the normal to $\mathcal{P}_2$ at R . (Do NOT prove this.)

- | | |
|--|----------|
| (iv) Find the values of t so that the distance between R and T is a minimum. | 2 |
|--|----------|

End of paper

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REFERENCE SHEET

– Mathematics –

– Mathematics Extension 1 –

– Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

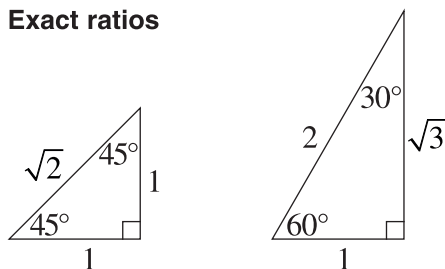
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$$

$$\cos(\theta + \phi) = \cos\theta\cos\phi - \sin\theta\sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta\tan\phi}$$

t formulae

If $t = \tan\frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

tangent: $y = tx - at^2$

normal: $x + ty = at^3 + 2at$

At (x_1, y_1) ,

tangent: $xx_1 = 2a(y + y_1)$

normal: $y - y_1 = -\frac{2a}{x_1}(x - x_1)$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$